

Optimization – Power Gating and DVFS

For my project optimization, I am implementing both Power Gating and Dynamic Voltage and Frequency Scaling (DVFS). Power gating will be used to reduce leakage by disabling idle blocks such as the multiplier, PRSG, and SISR using sleep transistors. DVFS will allow the system to dynamically scale down voltage and frequency during test mode and scale up during high-performance operations. Together, these techniques will support adaptive power management and serve as a foundation for my dissertation research on DVFS and low-power system design.